

DONNA Gluco-C Plus MSM Capsule

DESCRIPTION: Size 0 capsule, with red cap marked 'DONNA' and ivory body marked 'PLUS' with white to off white fill.

COMPOSITION Each capsule contains
Glucosamine Hydrochloride 300.87 mg equivalent to Glucosamine 250mg.
Chondroitin Sulphate Sodium 100mg
Methylsulfonylmethane (MSM) 150mg
Gelatin for hard capsule: Bovine source

ROUTE OF ADMINISTRATION: Oral

PHARMACODYNAMICS:

- (a) Arthrotrophic metabolic activity by stimulating the anabolic metabolism of osteo-cartilagineous tissues, through:
- stimulation of the biosynthesis of the mucopolysaccharides (which are the essential components of the cartilage ground-substance) and of the bone mesenchymal tissues in general.
- (b) Lubricant activity through:
- improvement of synovial fluid viscosity.
 - increase of synovial fluid production.

Glucosamine is recommended for the management of degenerative osteoarticular disease, eg. arthrosis and osteoarthritis both subacute and chronic with disappearance or reduction of articular pains, improvement of articular function, and inhibition or regression of the degenerative process.

Chondroitin, like glucosamine, slows the progression of arthritis. Chondroitin sulphate seems to protect the joints of osteoarthritis (OA) sufferers from further damage. Chondroitin sulphate is a complex carbohydrate that helps retain water.

Methylsulfonyl methane (MSM) is an organic sulphur-containing compound. It is an oxidation product of the organic solvent, dimethyl sulfoxide (DMSO). DMSO has been used in the treatment of arthritis and connective tissue injuries because cartilage has a high content of sulphur, and sulphur is needed for the formation of connective tissue. MSM as a source of sulphur could be useful in the management of conditions such as osteoarthritis and joint injuries where there is degeneration or destruction of cartilage.

PHARMACOKINETICS:

Pharmacokinetics of glucosamine in man: The pharmacokinetics of glucosamine sulphate was investigated in 6 healthy male volunteers (2 per administration route) using ¹⁴C uniformly labeled glucosamine sulphate and administering it in single dose by intravenous (i.v), intramuscular (i.m) or oral route. After oral administration a proportion dose to 90% of glucosamine sulphate is absorbed. Free glucosamine is not detectable in plasma during this study, because of the high detection limit of the method use. However, levels of Glucosamine Sulphate incorporated in plasma proteins reached a peak after 8-10 hours and then declined, exhibiting a half-life of 68 hours. Following oral administration, 10% was excreted via the urine, whereas 11.3% was found in the faeces. The radioactivity incorporated in the plasma proteins follows pharmacokinetics patterns which are similar to those after i.v. or i.m. administration, but its concentration in plasma is about 5 times smaller than that after parenteral administration. The AUC after oral administration is 26% of that after i.v., or i.m. administration. The plasma levels of radioactivity investigations in rats and dogs showing that also in man glucosamine sulphate is prodrug for glucosamine that is well absorbed after oral administration and that diffuses into several tissues, including bones and articular cartilages.

Absorption of chondroitin sulphate appears to occur from the stomach and small intestine. There is also an indication that some chondroitin sulphate, after absorption, does enter the joint space. Soft tissue distribution of radioactivity indicated a fairly homogeneous distribution throughout the body with relatively lower concentrations in skin and bone. Approximately 85.8% of the dose was recovered in the urine after 120 h, whereas only 3% was found in the faeces.

No quantifiable levels of radioactivity were found in any tissues after 120 h, indicating complete elimination of MSM. This indicates that MSM is rapidly absorbed, well distributed, and completely excreted from the body.

INDICATION: As an adjuvant therapy for osteoarthritis.

RECOMMENDED DOSAGE:

Mild to moderate osteoarthritis: 1 capsule twice daily for 6 weeks.
Severe osteoarthritis: 1 capsule three times a day for 8 weeks.
Maintenance dose: 1 capsule twice a day for 3-4 months.

CONTRAINDICATIONS: Contraindicated in patients hypersensitive to glucosamine and patients allergic to shellfish

WARNING & PRECAUTIONS:

Glucosamine may increase insulin resistance. Those with type 2 diabetes and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they consume glucosamine. Glucosamine and Chondroitin are not meant to replace NSAIDs in cases of intense pain as this is a causal therapy.

Children: The medical literature has not reported any adverse effects related to glucosamine use in children. Since young children may have undiagnosed allergies or medical conditions, glucosamine should not be used in children under 12 years of age unless recommended by a physician.
Glucosamine Hydrochloride is derived from seafood. Chondroitin Sulphate Sodium is derived from bovine source.



Black

Size: 130mm(W) x 195mm(H)

INTERACTION WITH OTHER MEDICATIONS:

An increased in International Normalised Ratio (INR) was noted when glucosamine was administered to patient who is treated with warfarin. Glucosamine may increase the anti-inflammatory activity of NSAIDS such as ibuprofen. This interaction may result in the need for lower doses of these medications. Those taking warfarin and those haemophilia should exercise caution in its use. Chondroitin sulphate should not be used concomitantly with chitosan. Glucosamine may increase insulin resistance and consequently affect glucose tolerance. It may reduce antidiabetic agent effectiveness.

eg from Micromedex :

- Severity: minor

- Onset: rapid

- Substantiation: theoretical

- Clinical Management: Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely.

- Probable Mechanism: Glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and/or alteration of peripheral glucose uptake.

Chitosan may form complexes with chondroitin sulfate decreasing its absorption.

USAGE IN PREGNANCY & LACTATION:

To date, there are no reported adverse effects related to fetal development during pregnancy or to infants who are breast-fed. However, it is recommended to consult healthcare practitioner if glucosamine is taken while pregnant or breastfeeding.

SIDE EFFECTS:

Cardiovascular: Peripheral oedema, tachycardia were reported in a few patients following a larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

Central nervous system: Drowsiness, headache, insomnia has been observed rarely during therapy (less than 1%).

Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.

Skin: Skin reactions such as erythema and pruritis have been reported with therapeutic administration of glucosamine.

SYMPTOMS AND TREATMENT OF OVERDOSE: In the event of an acute overdose, the patient should be observed closely and symptomatic treatment should be instituted.

STORAGE CONDITION: Store below 30°C. Protect from light. Keep out of reach of children. *Jauhi daripada kanak-kanak.*

PACK SIZE: Bottle of 30 and 90 capsules. Not all pack sizes are available locally.

SHELF LIFE: Please refer outer package

PRODUCT REGISTRATION HOLDER:

DUOPHARMA MARKETING SDN. BHD.
Lot No. 2,4,6,8 & 10, Jalan P/7, Section 13, Bangi Industrial Estate,
43650 Bandar Baru Bangi, Selangor, Malaysia.

MANUFACTURER :

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Reg. No.: MAL12025062X

Date of Revision: Apr 2024

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